

DATA VISUALIZATION & STORYTELLING



Insights from Bangalore Rapido Dataset

rapido

India's #1 Ride-hailing App

Quick, Affordable rides at your doorstep



Enter Pickup Location



OBJECTIVES

- **TO EXPLAIN THE DATASET AND TOOLS USED FOR ANALYSIS**
- **TO UNDERSTAND THE INSIGHTS REPRESENTED BY DIFFERENT DATA VISUALIZATIONS**



OBJECTIVES

- **TO INTERPRET TRENDS AND PATTERNS FROM CHARTS AND DASHBOARD, ANALYZE RIDE TRENDS**
- **TO EXPLAIN HOW DATA VISUALIZATION HELPS IN BETTER DECISION-MAKING**



INTRODUCTION

Data Visualization is the graphical representation of data using charts, graphs, and dashboards

**DATA
ANALYTICS?**



**DATA
VISUALIZATION?**

WHY RAPIDO USES DATA VISUALIZATION & DATA ANALYTICS



- To understand ride demand patterns across different times and locations
- To identify peak hours and allocate drivers efficiently
- To optimize pricing strategies based on demand and distance
- To analyze customer behavior and preferences
- To monitor ride completion and cancellation rates
- To improve operational efficiency and reduce waiting time
- To track revenue across different service types
- To support data-driven decision making
- To enhance customer experience and service quality

DATASET OVERVIEW

Bangalore Rapido Ride Services Dataset



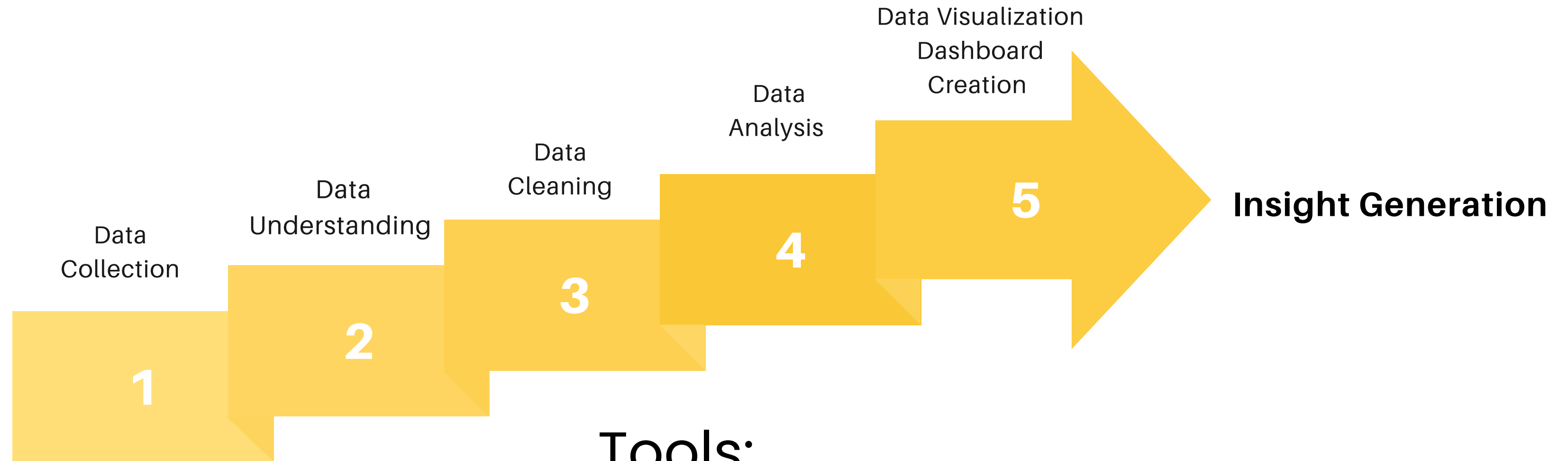
DATASET SOURCE: KAGGLE

NUMBER OF RECORDS: 50,000

NUMBER OF FEATURES:13

- **Numerical Data:**
duration, distance, ride_charge, misc_charge, total_fare
- **Categorical Data:**
services, ride_status, source, destination, payment_method, date, time, ride_id

WORKFLOW



Tools:

- Excel
- Kaggle
- Canva

DATA ANALYSIS & VISUALIZATION PROCESS

- Used Pivot Tables to summarize large data
- Created charts (bar, line, pie) for visualization
- Applied filters to analyze specific patterns
- Converted raw data into structured insights

Average of Total_Fare	Sum of Total_Fare	Count of Ride_ID
₹ 492	24612983.05	50,000

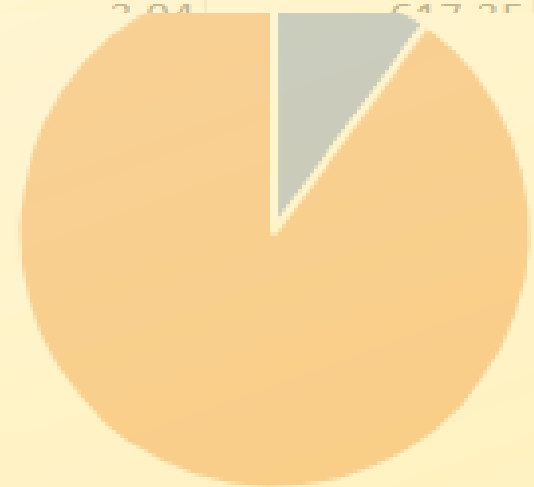
Row Labels	Sum of Revenue
auto	6099731.32
bike	7432783.95
bike lite	2387720.59
cab economy	5006233.04
parcel	3686514.15
Grand Total	24612983.05

Row Labels	Count of Ride_ID
Amazon Pay	11225
GPay	11268
Not Applicable	5036
Paytm	11315
QR scan	11156
Grand Total	50000

Row Labels	Count of Ride_ID
cancelled	5036
completed	44964
Grand Total	50000

DASHBOARD OVERVIEW

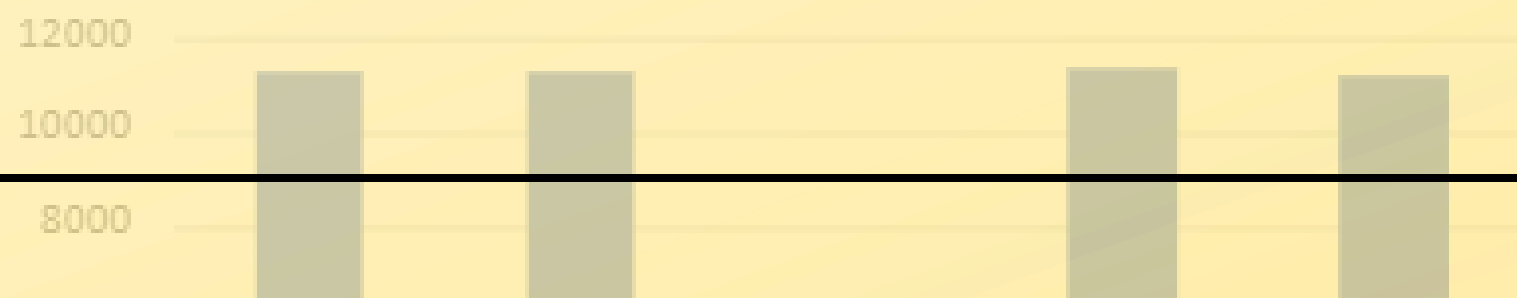
rides_data.xlsx



What will we understand from this dashboard?

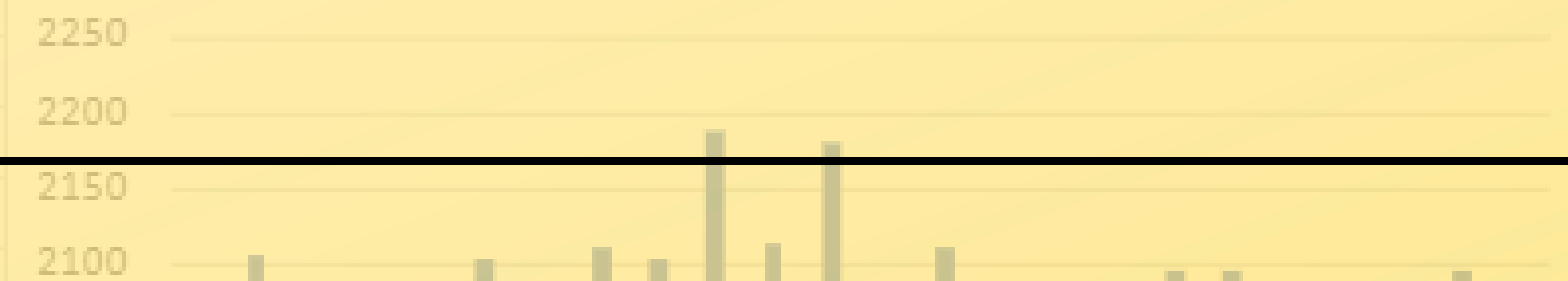
Count of Ride_ID

Payment Method Distribution



Count of Ride_ID

Ride Demand by Hour



 Total Rides

50,000

 Total Revenue

₹ 2.46 Cr

 Cancellation Rate

10.07%

 Avg Fare

₹ 492

Service_Type
auto
bike
bike lite
cab economy
parcel

Ride_Status
cancelled
completed



Rapido Ride Analytics Dashboard – Bangalore

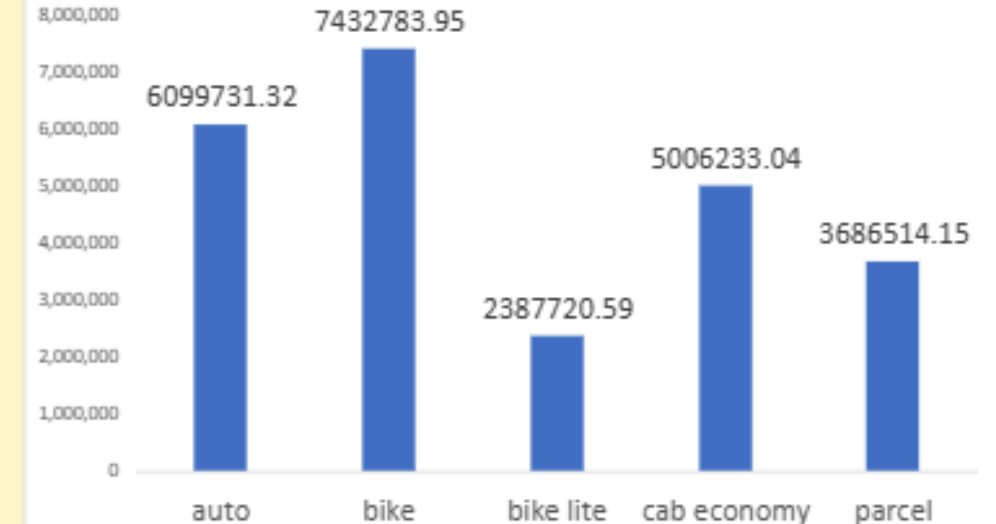
Monthly Ride Trends



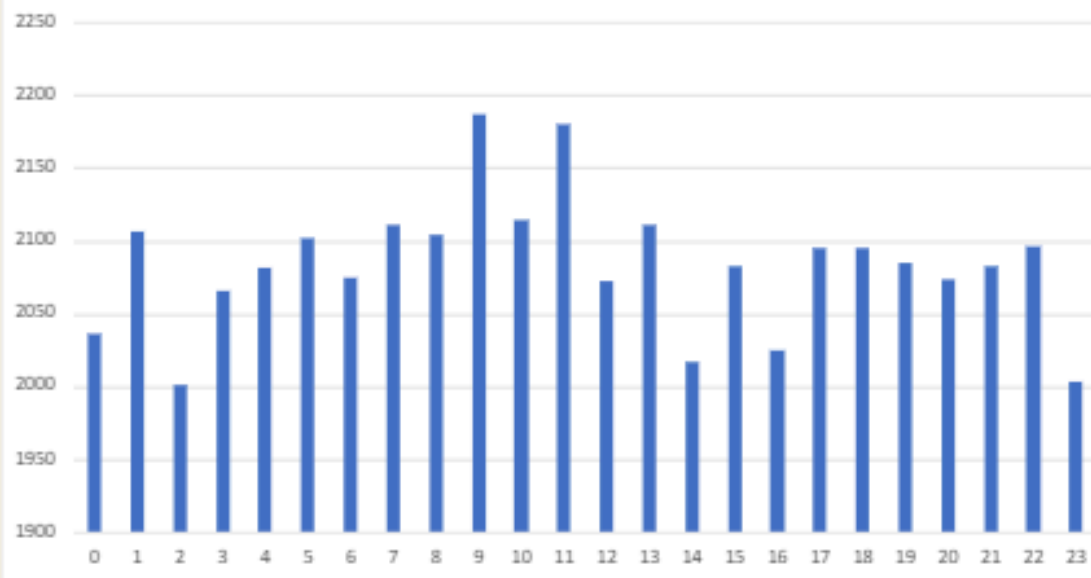
Ride Completion Status %



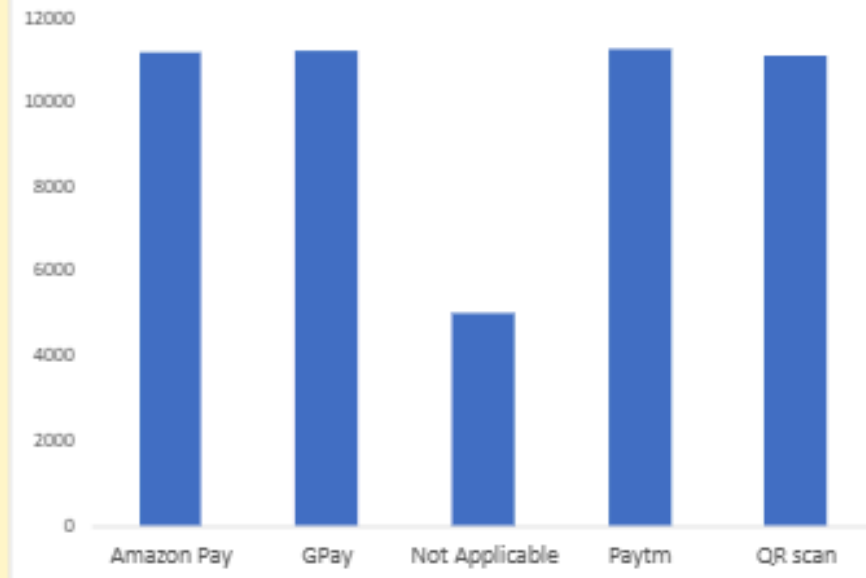
Revenue by Service Type



Hourly Ride Demand



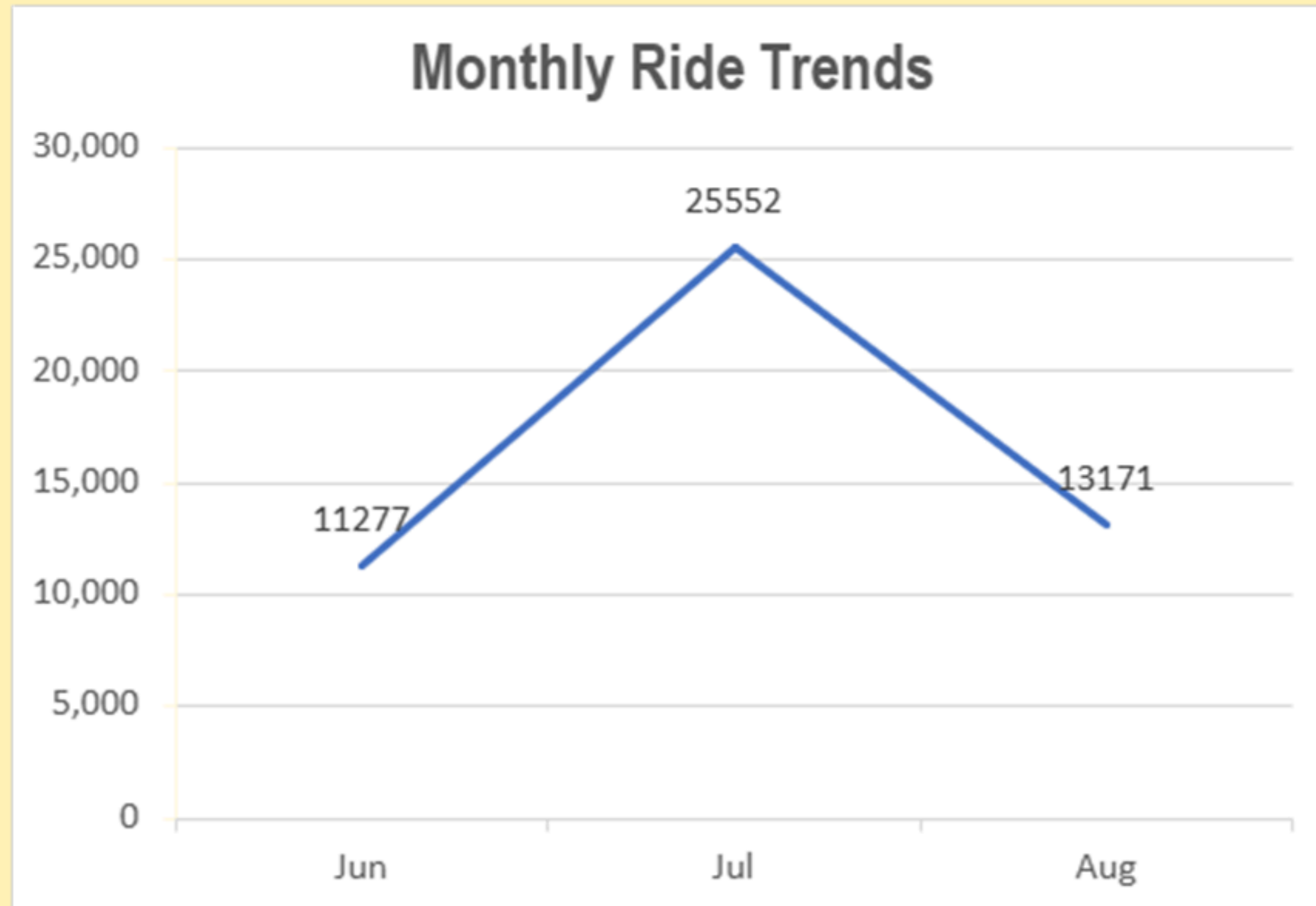
Payment Method Usage



Ride_Day
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday

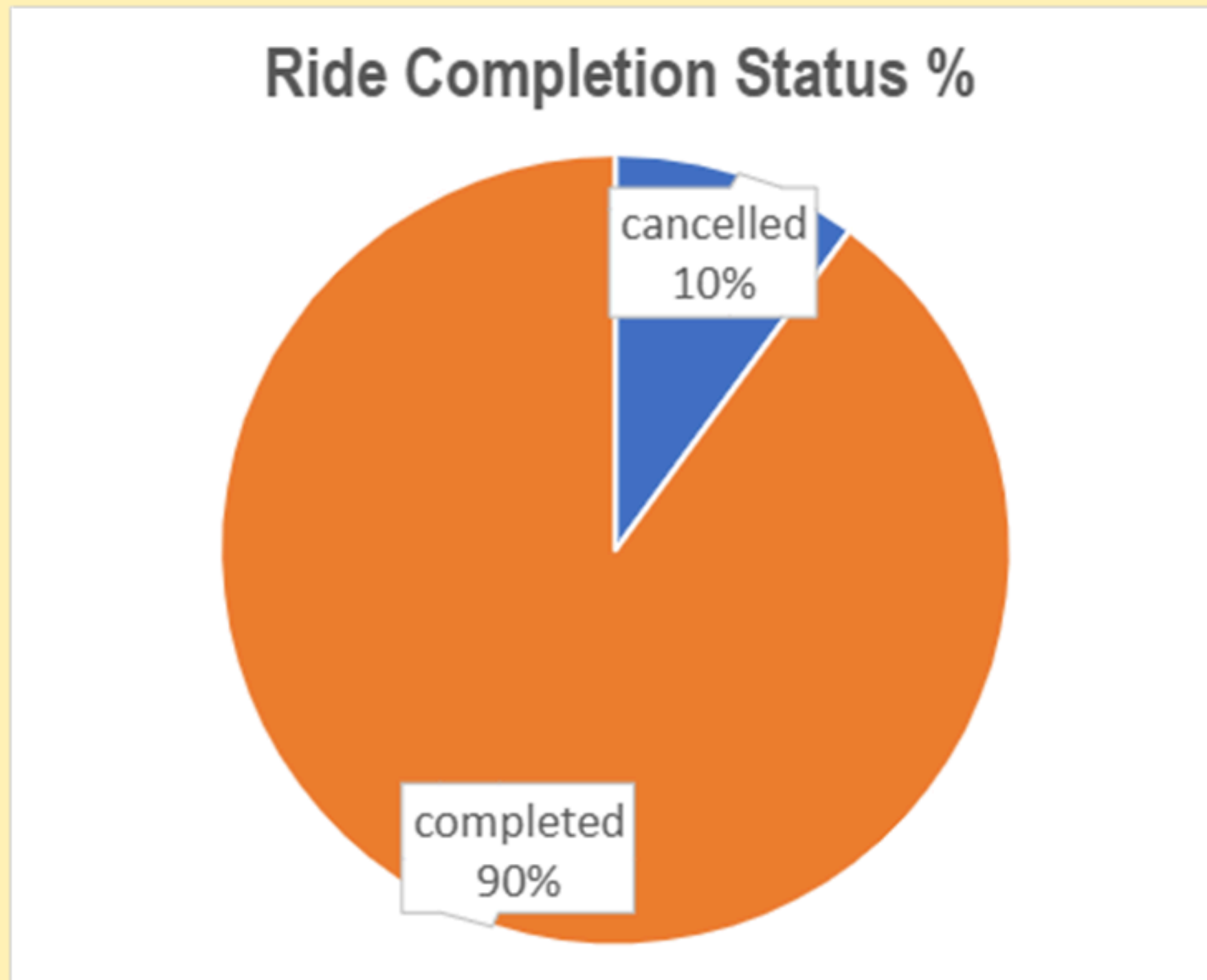
Payment_Method
Amazon Pay
GPay
Not Applicable
Paytm
QR scan

1. RIDE TRENDS OVER TIME (LINE CHART)



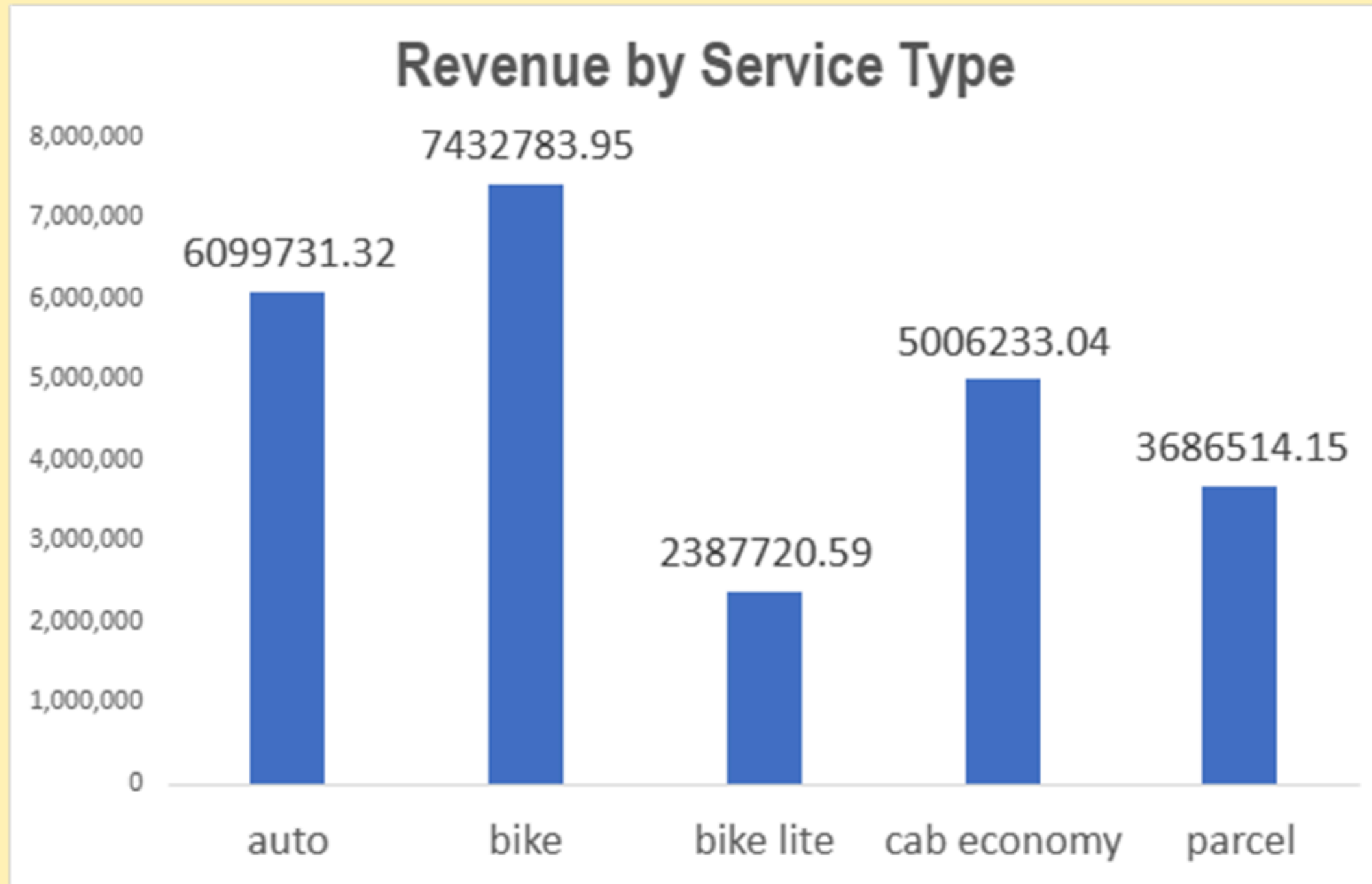
- TIME-SERIES ANALYSIS
- HELPS IDENTIFY TRENDS OVER TIME

2. RIDE COMPLETION STATUS (PIE CHART)



- CATEGORICAL DATA DISTRIBUTION
- HELPS UNDERSTAND PROPORTIONS

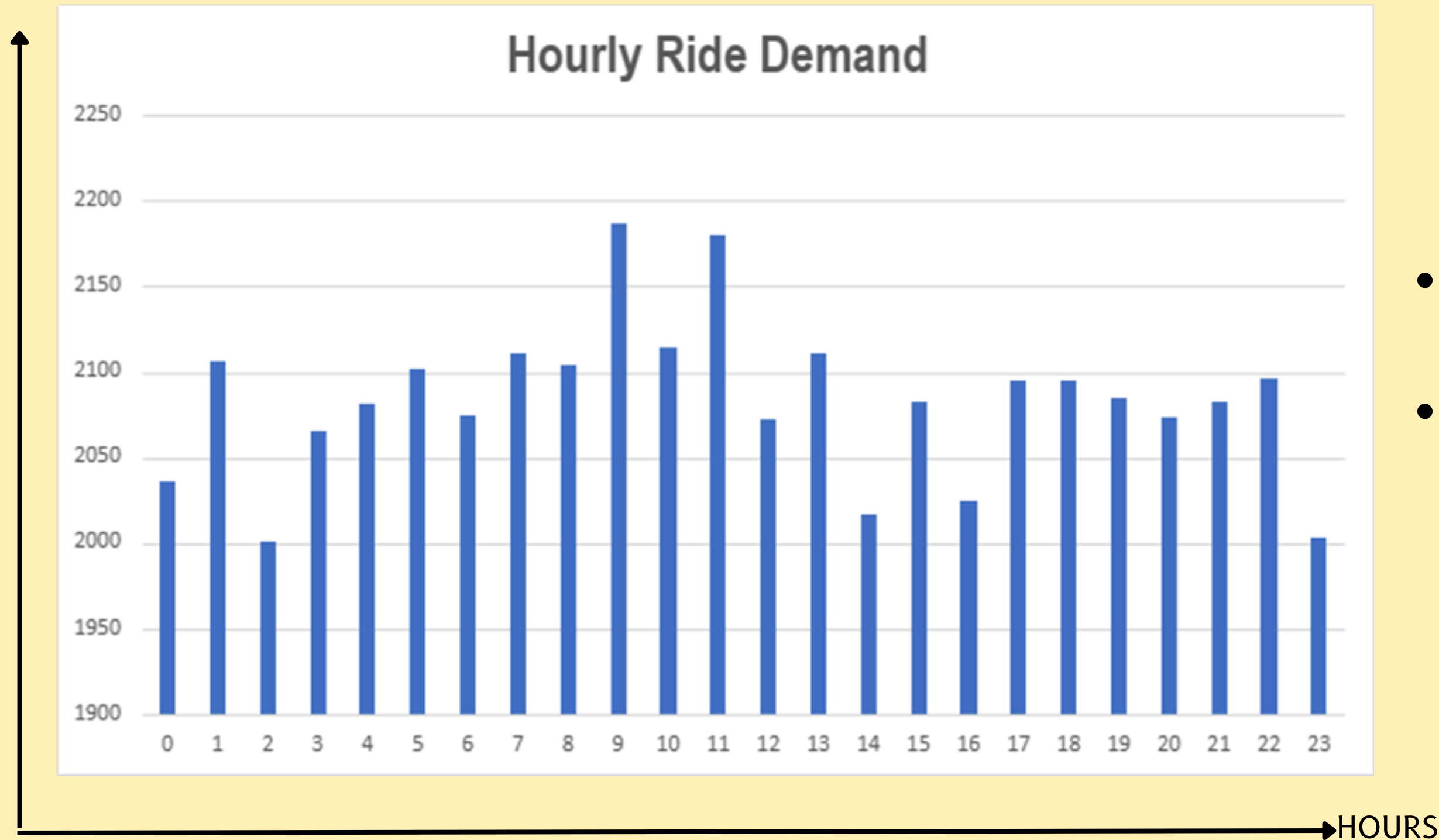
3. REVENUE BY SERVICE TYPE (BAR CHART)



- COMPARISON USING BAR CHARTS [EDA]
- HELPS IDENTIFY WHICH CATEGORY PERFORMS BEST

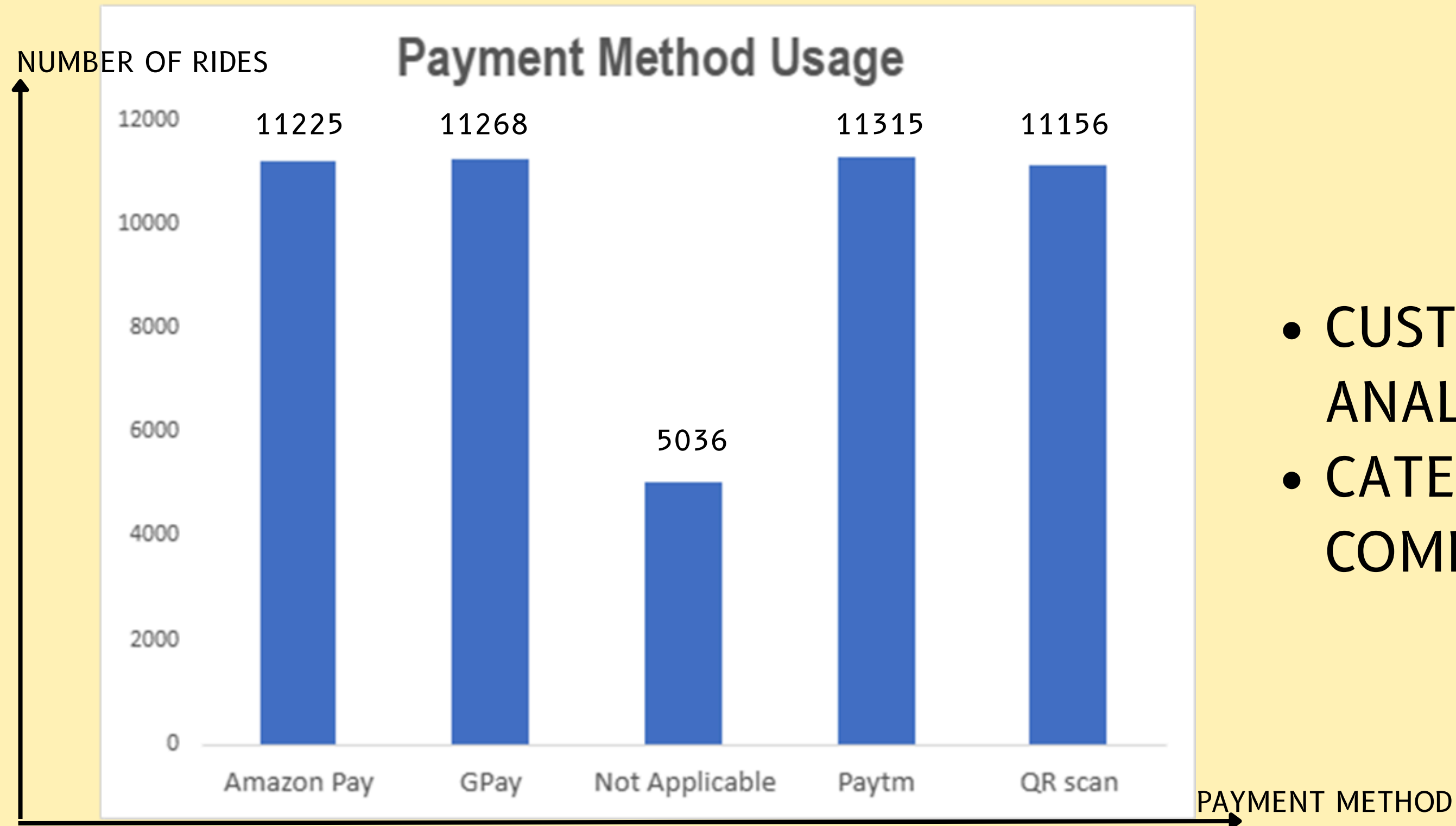
4. RIDE DEMAND BY HOUR (BAR CHART)

NUMBER OF RIDES



- PATTERN RECOGNITION IN DATA
- TIME-BASED ANALYSIS

5. PAYMENT METHOD DISTRIBUTION



- CUSTOMER BEHAVIOR ANALYSIS
- CATEGORICAL COMPARISON

6. KEY PERFORMANCE INDICATORS

 Total Rides
50,000

 Total Revenue
₹ 2.46 Cr

 Cancellation Rate
10.07%

 Avg Fare
₹ 492

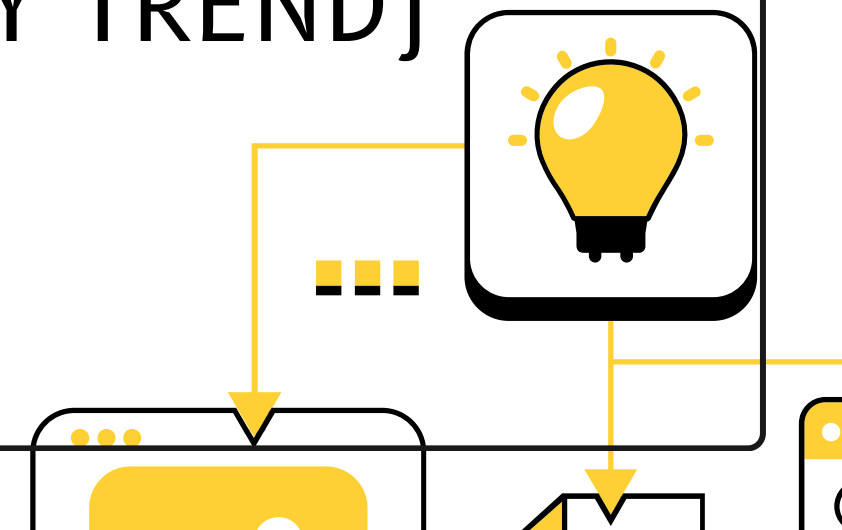
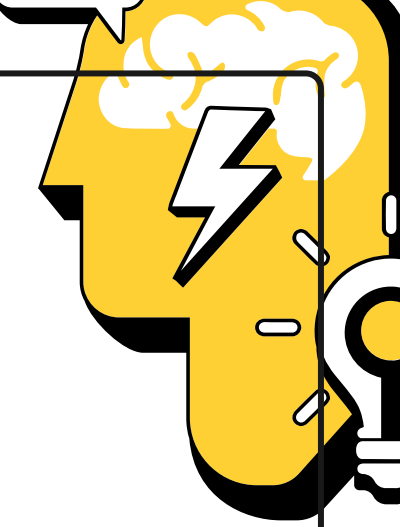
Service_Type
auto
bike
bike lite
cab economy
parcel

Ride_Status
cancelled
completed

- TOTAL RIDES: TOTAL NUMBER OF RIDES COMPLETED IN THE DATASET
- TOTAL REVENUE: OVERALL REVENUE GENERATED FROM ALL RIDES
- AVERAGE FARE: AVERAGE COST PER RIDE
- CANCELLATION RATE [%]: PERCENTAGE OF RIDES CANCELLED
- PEAK RIDE HOUR: TIME PERIOD WITH HIGHEST RIDE DEMAND
- TOP SERVICE TYPE: SERVICE GENERATING HIGHEST REVENUE (E.G., BIKE)

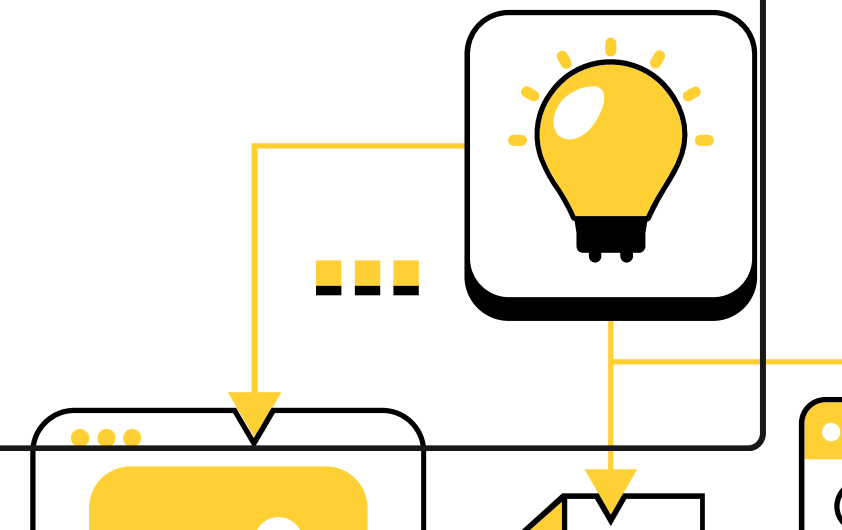
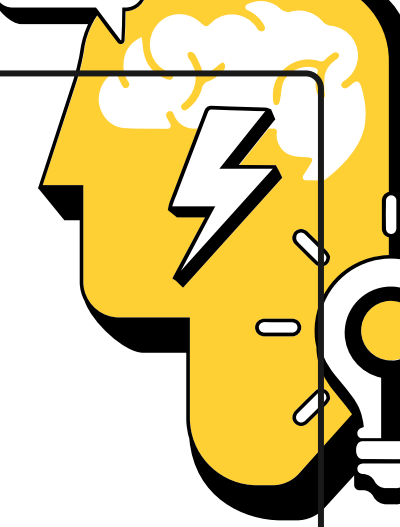
KEY INSIGHTS

- PEAK RIDE DEMAND OCCURS DURING MORNING AND EVENING HOURS
- BIKE SERVICES GENERATE HIGHEST REVENUE AMONG ALL SERVICE TYPES
- LOW CANCELLATION RATE [$\sim 10\%$] INDICATES EFFICIENT SERVICE
- DIGITAL PAYMENTS ARE HIGHLY PREFERRED BY USERS
- RIDE DEMAND SHOWS VARIATION OVER TIME [MONTHLY TREND]



BUSINESS INSIGHTS / RECOMMENDATIONS

- INCREASE BIKE AVAILABILITY DURING PEAK HOURS
- OPTIMIZE PRICING STRATEGIES DURING HIGH DEMAND PERIODS
- FOCUS ON HIGH-DEMAND LOCATIONS FOR BETTER SERVICE COVERAGE
- ENCOURAGE DIGITAL PAYMENTS FOR FASTER TRANSACTIONS
- REDUCE CANCELLATIONS BY IMPROVING DRIVER ALLOCATION AND MATCHING



[Overviews](#)[Customers](#)[Captains](#)

WHY THESE INSIGHTS

Safety for all

ARE USEFUL?

At Rapido, the well-being of our customers is above everything else. We are constantly in pursuit of enhancing our safety measures to ensure every Rapido ride is a pleasant and comfortable experience.



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CONCLUSION (AFTER DATA VISUALIZATION)

- Ride demand is highest during peak hours [morning & evening]
- Bike services dominate in terms of usage and revenue
- Majority of rides are successfully completed [low cancellations]
- Users show strong preference for digital payment methods
- Demand varies across time and locations, indicating usage patterns



THANK YOU

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